مسار

برنامج تدريب ميداني

2025

Introduction to Masar Program

Agenda

- Introduction into Cybersecurity
- Cybersecurity Roles
- Program Outline

Introduction to Cybersecurity

Introduction to Cybersecurity

- **Cybersecurity** is the practice of protecting digital assets, from unauthorized access, disruption, or destruction
- This protection can be:
  - Technological security control
  - Moral control such as raising security awareness
  - Proactive security approach such as penetration testing
- **Cyber attacks** are malicious actions performed by threat actors in order to perform harmful actions, whether damaging, destroying or stealing digital data and system.

Introduction to Cybersecurity

- Threat actors performing cyber attack are usually broken down to:
  - **Nation-State Actors** – Government-backed groups conducting espionage, sabotage, or disruption.
  - **Cybercriminals** – Profit-driven actors seeking financial gain through fraud, ransomware, or theft.
  - **Hacktivists** – Ideologically motivated groups targeting organizations to promote political or social causes.
  - **Insiders** – Employees, contractors, or partners misusing authorized access (maliciously or negligently).
  - **Script Kiddies** – Inexperienced individuals using readily available tools to launch attacks with little technical skill.

Cybersecurity Roles

Cybersecurity Roles

- **Offensive Cybersecurity**
- **Defensive Cybersecurity**
- **Governance, Risk Management, and Compliance**

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Cybersecurity Roles

- **Offensive Security Engineers** simulate attacks against systems, networks, websites, and digital assets in order to uncover the weaknesses and vulnerabilities, they use the same techniques used by threat actors and hackers to ensure visibility of the security posture for an organization, they eventually advice for the proper security controls and mitigations.

Penetration Tester	Assess and identifies vulnerabilities in computer systems, networks, or applications by simulating real-world cyberattacks to strengthen security defenses
Red Teamer	Simulates adversarial threat actors by executing realistic and sophisticated attacks on an organization's systems and processes
Application Security Engineer	Ensures the security of software applications by collaborating with software developers, by identifying, assessing and mitigating software security vulnerabilities

Cybersecurity Roles

- **Defensive Security Engineers** are responsible for safeguarding an organization's systems, networks, and digital assets against potential cyber threats and attacks.
- They focus on developing and implementing security measures to protect against real-world threats, by designing, implementing, and managing security controls.
- They monitor security incidents, and work proactively to prevent and mitigate potential security breaches to strengthen the overall security posture of an organization.

Cybersecurity Roles

- **Defensive Security Engineers** can be divided into different roles with different responsibilities

SOC Analyst	Monitors and responds to security incidents, by managing security tools, and ensuring the overall security posture.
Incident Responder (CSIRT)	Investigating breaches, coordinating responses, and implementing measures to prevent future occurrences.
Digital Forensics Analyst (CSIRT)	Utilizes special techniques and tools to investigate and analyze digital evidence, such as computer systems, storage devices, and networks, to support legal investigations and resolve cybersecurity incidents.
Security Engineer	Designs and implements security measures and controls within an organization's IT infrastructure, ensuring the integrity, confidentiality, and availability of systems and data by creating comprehensive security strategies and solutions.
Threat Analyst	Gathers and analyzes external intelligence to enrich detections and predict adversary activity.

Cybersecurity Roles

- **GRC** are essential components of a strong cybersecurity strategy.
  - **Governance** sets strategic direction, defines roles, and allocates resources.
  - **Risk** Management prioritizes threats for informed decision-making.
  - **Compliance** ensures adherence to standards, reducing legal and financial risks.
- Together, these form a comprehensive and proactive cybersecurity framework, safeguarding against cyber threats while maintaining regulatory compliance.

Cybersecurity Roles

- **Governance, Risk Management, and Compliance** can be divided into different roles with different responsibilities

Governance Manager	Governance Managers are tasked with developing and enforcing policies, processes, and procedures that guide the organization's cybersecurity strategy. They establish frameworks for decision-making, define roles and responsibilities, and ensure that cybersecurity aligns with the organization's overall governance structure.
Risk Manager	Risk Managers focus on identifying, assessing, and prioritizing potential cybersecurity risks. They work to develop risk mitigation strategies and ensure that the organization's resources are allocated efficiently to address the most critical threats.
Compliance Officer	Compliance Officers are responsible for ensuring that the organization adheres to all relevant laws, regulations, and internal policies. They conduct regular audits and assessments to verify compliance, create and update compliance documentation, and liaise with regulatory bodies as needed.
Security Auditor	Security Auditors evaluate the effectiveness of an organization's security controls and measures. They conduct thorough assessments to identify vulnerabilities and weaknesses, ensuring that the organization's cybersecurity practices are in line with industry standards and best practices.

Program Outline

Program Prerequisites

- **1. Personal Computer Requirements:**
  - Processor: Intel i5 (8th Gen or later) / AMD Ryzen 5 or equivalent
  - Memory (RAM): Minimum 8 GB (16 GB recommended)
  - Storage: At least 50 GB of free disk space
  - Operating System: Windows 10 / 11, macOS, or Linux (latest updates installed)
  - Virtualization Support: System must support and have enabled hardware virtualization (VT-x / AMD-V) in BIOS to run virtual machines.
- **2. Internet Connection:**
  - Stable broadband connection with at least 10 Mbps download speed (25 Mbps recommended).
  - Ability to access and stream video conferencing sessions without interruption.

Program Outline

Weeks	Topics	Sub Topics
1	Introduction to Masar Program	• Introduction into Cybersecurity • Cybersecurity Roles • Program Outline • Introduction to Linux
2-4	Offensive Security	• Introduction to Hacking Methodology • Reconnaissance & Scanning • Vulnerability Assessment • Exploitation • Linux Post Exploitation • Clean up, Reporting & C2 • Secure Coding Foundations • Application Security Vulnerabilities • DevSecOps

Program Outline

Weeks	Topics	Sub Topics
5-7	Defensive Security	• Introduction to Blue Team • Incident Handling & Response • Digital Forensics • Secure Configuration • Threat Intelligence and Threat Hunting
8	Governance, Risk & Compliance	• Risk Assessment and Management • Introduction to Business Continuity • Introduction to Disaster Recovery

Program Practical Outline

- **Exercises:** practical exercises designed to apply theoretical knowledge to real-world scenarios or tasks in each session
- You'll complete a new exercise each day, with your instructor guiding you on whether it will be an individual or group activity
- **Take-home Assignments:** are extended tasks you complete outside of class to apply what you've learned in real-world or project-based scenarios, allowing deeper exploration and independent problem-solving

Structure of Training

~200 instructional hours over 8 weeks:

Detail	Value
Duration	8 Weeks
Sessions per Week	3 Days (Online)
Hours per Day	7 hours (including 1 hour break)
Take-Home Assignments (Daily)	2.5 hours
Total Instructional Sessions	24 (3 days/week × 8 weeks)
Total Hours per Student	204 hours (including take-home tasks)

Sample Training Day Schedule

Time	Activity	Duration
09:00 – 11:30	Theoretical Session	2.5 hrs
11:30 – 12:30	Lunch Break	1 hr
12:30 – 2:00	Theoretical Session	1.5 hrs
2:00 – 4:00	In-Session Exercise	2 hrs

Grading & Assessment

Component	Weight
Attendance	30%
In-Class Tasks & Projects	40%
Take-Home Assignments	30%

Program Platforms

- **Webinar**: Our online meeting hosting platform at https://live.zoho.com/weow-xgv-gsh
- **Learning Management System:** Access the training material, exercises and take-home assignments at https://training.skidz.io/lms/courses/masar-program/

Program Exercise Fictional Scenario

- **Bazaarjo** is the largest e-commerce platform that enables merchants to list their products and allow end users to purchase them online, with a specific focus on the bazaar market.

Program Exercise Fictional Scenario Part 1

- The **National Cyber Security Center** in Jordan contacted Bazaarjo's CTO with a serious warning: their threat intelligence team had discovered secret customer information, including usernames, passwords, and personal data, being traded on the dark web!

- The breach began with a simple phishing attack.
- A backend developer named Omar received an email asking him to download a new HR system desktop application to update his employee details, such as his graduation year.
- Believing it to be legitimate, Omar installed the program. In reality, it was malware called Lumma.
- Once running on his computer, it stole all of his saved credentials, especially those stored in his browser.
- Among them were the login details for Bazaarjo's source code management platform.

Program Exercise Fictional Scenario Part 2

- With this access, the attackers added malicious code directly into the company's repositories.
- When this code was executed by the CI/CD runners, it gave the attackers a pathway into Bazaarjo's cloud infrastructure and internal systems.
- From there, they reached production servers and application databases, stealing the entire set of customer records.
- Eventually, they decided to sell this stolen data on the dark web.

Program Code of Conduct and Attendance Policy

- **Code of Conduct:** a set of rules, principles, and guidelines that outline expected and acceptable behavior within an organization.
- You can access the code of conduct at here.
- **Attendance Policy:** Attendance will be recorded three times during the day through links shared at random intervals throughout the session.

Q&A